



10855 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI imaging F770W+F1130W+F1800W+F2100W+F2550W	MIRI Imaging	(1) J1107
	2	NIRCam Imaging F480M + F187N	NIRCam Imaging	(1) J1107
	3	MIRI imaging F770W+F1130W+F1800W+F2100W+F2550W	MIRI Imaging	(2) J1244
	4	NIRCam Imaging F480M + F187N + F115W + F410M	NIRCam Imaging	(2) J1244

Folder	Observation	Label	Observing Template	Science Target
	5	MIRI imaging F770W+F1130W+F1500W+F2100W+F2550W	MIRI Imaging	(3) J1613
	6	NIRCam Imaging F480M + F187N	NIRCam Imaging	(3) J1613
	7	MIRI imaging F770W+F1130W+F1500W+F2100W+F2550W	MIRI Imaging	(4) J1622
	8	NIRCam Imaging F480M + F187N + F115W + F410M	NIRCam Imaging	(4) J1622

ABSTRACT

The recent JWST discovery of warm circumgalactic dust in the 100-kpc wind of Makani, a massive compact starburst galaxy at $z = 0.459$, provides the strongest evidence to date that the dust entrained in galactic winds survives the long journey to the CGM, but is eroded along the way. Dusty CGM-scale winds, if common in the early universe, may help explain the large amount of dust outside of galaxies, inferred from reddening measurements of background quasars and galaxies by foreground galaxy halos. We propose to study four additional massive starburst galaxies that have CGM-scale winds and span a representative range of starburst and wind properties. These objects are selected to be at the same distance as Makani to take advantage of a match between the bandpasses of the NIRCam/MIRI filters and redshifted Pa-beta, PAH 3.3, 7.7, and 11.3. These data will be compared with published multi-phase data cubes of these winds to (1) provide a census of the warm dust across the winds and identify the driving parameter(s) of its distribution, (2) explore whether the dust evolution observed in Makani is common or sensitive on the outflow timescale or kinematics (e.g. shocks), and (3) provide a holistic view of the dust-gas cycle in these galaxy-CGM ecosystems in formation. The warm dust traced by JWST fills the temperature gap (100 - 1000 K) between the observed cool clouds and warm ionized gas. If the dust is to survive to large distances then it must be shielded against thermal sputtering and the many other processes taking place in the harsh environment of galactic winds, perhaps via cloud-wind mixing and condensation. These data will inform modeling of these complex processes.

OBSERVING DESCRIPTION

We image the CGM of four galaxies using MIRI and NIRCam with the following settings:

-MIRI imaging

acquisition: point and shoot

JWST Proposal 10855 (Created: Friday, May 15, 2026, 12:00:18PM Eastern Standard Time) - Overview

FULL array

4-point extended source dither pattern

Readout mode: FAST

Filters F1000W, F1500W (two $z < 0.45$ objects) or F1800W (two $z > 0.45$ objects), and F2100W
(sequentially, in order of increasing wavelength to avoid persistence)

F770W: 54 groups x 5 exposures per dither position x 4 dither positions

--> total exposure time on source: 51 min

F1130W: 38 groups x 7 exposures per dither position x 4 dither positions

--> total exposure time on source: 50 min

F1500W or F1800W: 29 groups x 9 exposures per dither position x 4 dither positions

--> total exposure time on source: 50 min

F2100W: 28 groups x 11 exposures per dither position x 4 dither positions

--> total exposure time on source: 59 min

No off-source background images

-NIRCam imaging

acquisition: point and shoot

Module B only

SUB400P subarray

No primary dither, STANDARD subpixel dither, 4 subpixel positions

Readout mode: MEDIUM8

Filters F480M (long) and F187N (short) simultaneously

10 groups x 5 integrations per dither position x 4 dither positions

--> total exposure time on source: 55 min

No off-source background images

For J1244 and J1622, we also obtain NIRCam "snapshots" at 1.15 μm (and 4 μm for free) to match the depth of the existing HST/WFC3 F814W images of the other three galaxies in the sample:

JWST Proposal 10855 (Created: Friday, May 15, 2026, 12:00:18PM Eastern Standard Time) - Overview

-NIRCam imaging

acquisition: point and shoot

Module B only

SUB400P subarray

No primary dither, STANDARD subpixel dither, 4 subpixel positions

Readout mode: MEDIUM8

Filters F115W (short) and F410M (long) simultaneously

8 groups x 1 integration per dither position x 4 dither positions

--> total exposure time on source: 9 min

No off-source background images

Proposal 10855 - Targets - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	J1107	RA: 11 07 2.8632 (166.7619300d) Dec: +04 17 2.47 (4.28402d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the NED database.</i> <i>Category=Galaxy</i> <i>Description=[Starburst galaxies]</i> <i>Extended=YES</i>		
	(2)	J1244	RA: 12 44 21.7056 (191.0904400d) Dec: +41 40 28.96 (41.67471d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the NED database.</i> <i>Category=Galaxy</i> <i>Description=[Starburst galaxies]</i> <i>Extended=YES</i>		
	(3)	J1613	RA: 16 13 32.5200 (243.3855000d) Dec: +28 34 14.52 (28.57070d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the NED database.</i> <i>Category=Galaxy</i> <i>Description=[Starburst galaxies]</i> <i>Extended=YES</i>		
	(4)	J1622	RA: 16 22 47.1072 (245.6962800d) Dec: +31 45 32.87 (31.75913d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the NED database.</i> <i>Category=Galaxy</i> <i>Description=[Starburst galaxies]</i> <i>Extended=YES</i>		

Proposal 10855 - Observation 1 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 1: MIRI imaging F770W+F1130W+F1800W+F2100W+F2550W										Fri May 15 17:00:19 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	J1107	RA: 11 07 2.8632 (166.7619300d) Dec: +04 17 2.47 (4.28402d) Equinox: J2000								
Template	Comments: This object was generated by the targetselector and retrieved from the NED database.										
	Category=Galaxy Description=[Starburst galaxies] Extended=YES										
Dithers	Subarray										
	FULL										
Spectral Elements	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	54	5	1	Dither 1	4	20	3041.444	269889
	2	F1130W	FASTR1	38	7	1	Dither 1	4	28	3019.244	269889
	3	F1800W	FASTR1	29	9	1	Dither 1	4	36	2985.943	269889
	4	F2100W	FASTR1	28	11	1	Dither 1	4	44	3529.851	269889

Proposal 10855 - Observation 2 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 2: NIRCam Imaging F480M + F187N										Fri May 15 17:00:19 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	J1107	RA: 11 07 2.8632 (166.7619300d) Dec: +04 17 2.47 (4.28402d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the NED database.</i> <i>Category=Galaxy</i> <i>Description=[Starburst galaxies]</i> <i>Extended=YES</i>								
Template	Module					Subarray					
	B					SUB400P					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F187N	F480M	MEDIUM8	10	5	20	4	3279.765	269889	
Special Requirements	Aperture PA Range 0.3025525 to 63.3025525 Degrees (V3 0.0 to 63.0) Aperture PA Range 79.3025525 to 359.3025525 Degrees (V3 79.0 to 359.0)										

Proposal 10855 - Observation 3 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 3: MIRI imaging F770W+F1130W+F1800W+F2100W+F2550W										Fri May 15 17:00:19 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	J1244	RA: 12 44 21.7056 (191.0904400d)								
			Dec: +41 40 28.96 (41.67471d)								
			Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the NED database.										
	Category=Galaxy										
	Description=[Starburst galaxies]										
	Extended=YES										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	54	5	1	Dither 1	4	20	3041.444	269889
	2	F1130W	FASTR1	38	7	1	Dither 1	4	28	3019.244	269889
	3	F1800W	FASTR1	29	9	1	Dither 1	4	36	2985.943	269889
	4	F2100W	FASTR1	28	11	1	Dither 1	4	44	3529.851	269889

Proposal 10855 - Observation 4 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 4: NIRCcam Imaging F480M + F187N + F115W + F410M									Fri May 15 17:00:19 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCcam Imaging									
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(2)	J1244	RA: 12 44 21.7056 (191.0904400d)							
			Dec: +41 40 28.96 (41.67471d)							
			Equinox: J2000							
			Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Starburst galaxies] Extended=YES							
Template	Module					Subarray				
	B					SUB400P				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				4
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F187N	F480M	MEDIUM8	10	6	24	4	3935.718	269889
	2	F115W	F410M	MEDIUM8	8	1	4	4	523.454	269889

Proposal 10855 - Observation 5 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 5: MIRI imaging F770W+F1130W+F1500W+F2100W+F2550W										Fri May 15 17:00:19 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	J1613	RA: 16 13 32.5200 (243.3855000d) Dec: +28 34 14.52 (28.57070d) Equinox: J2000								
Template	Comments: This object was generated by the targetselector and retrieved from the NED database.										
	Category=Galaxy Description=[Starburst galaxies] Extended=YES										
Dithers	Subarray										
	FULL										
Spectral Elements	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	54	5	1	Dither 1	4	20	3041.444	269889
	2	F1130W	FASTR1	38	7	1	Dither 1	4	28	3019.244	269889
	3	F1500W	FASTR1	29	9	1	Dither 1	4	36	2985.943	269889
	4	F2100W	FASTR1	28	11	1	Dither 1	4	44	3529.851	269889

Proposal 10855 - Observation 6 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 6: NIRCam Imaging F480M + F187N									Fri May 15 17:00:19 GMT 2026	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	J1613	RA: 16 13 32.5200 (243.3855000d)								
			Dec: +28 34 14.52 (28.57070d)								
			Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the NED database.										
	Category=Galaxy										
	Description=[Starburst galaxies]										
	Extended=YES										
Template	Module					Subarray					
	B					SUB400P					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F187N	F480M	MEDIUM8	10	5	20	4	3279.765	269889	
Special Requirements	Aperture PA Range 0.3025525 to 149.3025525 Degrees (V3 0.0 to 149.0)										
	Aperture PA Range 167.3025525 to 359.3025525 Degrees (V3 167.0 to 359.0)										

Proposal 10855 - Observation 7 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 7: MIRI imaging F770W+F1130W+F1500W+F2100W+F2550W										Fri May 15 17:00:19 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	J1622	RA: 16 22 47.1072 (245.6962800d) Dec: +31 45 32.87 (31.75913d) Equinox: J2000								
Template	Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Starburst galaxies] Extended=YES										
	Subarray										
Dithers	FULL										
	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
Spectral Elements	1	CYCLING	1	4		5	1			LARGE	
	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	54	5	1	Dither 1	4	20	3041.444	269889
	2	F1130W	FASTR1	38	7	1	Dither 1	4	28	3019.244	269889
	3	F1500W	FASTR1	29	9	1	Dither 1	4	36	2985.943	269889
4	F2100W	FASTR1	28	11	1	Dither 1	4	44	3529.851	269889	

Proposal 10855 - Observation 8 - Deep Imaging Survey of the Warm Circumgalactic Dust around Starburst Galaxies

Observation	Proposal 10855, Observation 8: NIRCam Imaging F480M + F187N + F115W + F410M									Fri May 15 17:00:19 GMT 2026	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	J1622	RA: 16 22 47.1072 (245.6962800d)								
			Dec: +31 45 32.87 (31.75913d)								
			Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the NED database.										
	Category=Galaxy										
	Description=[Starburst galaxies]										
	Extended=YES										
Template	Module					Subarray					
	B					SUB400P					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F187N	F480M	MEDIUM8	9	6	24	4	3538.22	269889	
	2	F115W	F410M	MEDIUM8	8	1	4	4	523.454	269889	
Special Requirements	Aperture PA Range 11.3025525 to 156.3025525 Degrees (V3 11.0 to 156.0) Aperture PA Range 165.3025525 to 350.3025525 Degrees (V3 165.0 to 350.0)										